

Lilian William Hollard

Ph.D. Candidate in Machine Learning

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🌐 LinkedIn

🌐 Portfolio

Professional Summary

Ph.D. researcher in Machine Learning with over 3 years of experience in optimizing deep neural network architectures, specializing in CNNs, Transformers, real-time object detection, and image classification. Proven track record in reducing model complexity while maintaining high accuracy for industrial applications.

Education

- **University of Reims Champagne Ardenne** *Reims, France*
Ph.D. in Machine Learning, Lightweight Vision Models for Embedded Systems Feb. 2023 – Expected Nov. 2025
- **University of Reims Champagne Ardenne** *Reims, France*
Master of Science in Computer Science Sep. 2020 – June 2022

Professional Experience

- **LICIIS Laboratory** *Reims, France*
Machine Learning Researcher Nov. 2022 – Present
 - **Neural Network Architecture Design:** Develop, implement, and optimize deep neural networks for Computer Vision applications, achieving significant cost reduction while improving accuracy using PyTorch framework.
 - **CNN Performance Optimization:** Reduced YOLO model inference time by 26% for industrial applications with zero accuracy loss compared to state-of-the-art object detectors.
 - **Transformer Architecture Innovation:** Engineered one of the lightest Vision Transformers for classification (0.9M parameters, 70% ImageNet accuracy), optimizing costly architectures for resource-constrained environments.
 - **Infrastructure Development:** Built comprehensive analytics environment using Docker, AWS, and Romeo Cloud; authored core Computer Vision libraries adopted across research teams.
 - **Research Publications:** Authored multiple first-author papers in English and French for peer-reviewed international conferences.
 - **Project Leadership:** Managed complete ML lifecycle from raw data processing to deployment-ready model implementation.
 - **AI Production Systems:** Designed and implemented AI pipelines for vineyard yield estimation, significantly improving harvest efficiency through precision agriculture.
 - **Grant Acquisition & Strategy:** Contributed to proposal writing, funding acquisition, and EU grant documentation for multi-million euro research projects.
 - **Applied Computer Vision:** Developed real-time object detection systems for vineyard monitoring, enabling embedded deployment in agricultural robotics.
 - **Industry Collaboration:** Facilitated technical alignment between academic researchers and industry stakeholders through regular workshops and demonstrations.
- **University of Reims Champagne Ardenne (URCA)** *Reims, France*
Associate Professor Sep. 2024 – April 2025
 - **Advanced Teaching:** Delivered graduate-level Deep Learning courses, designed practical neural network implementations in PyTorch, and organized hands-on model training workshops.
 - **Teaching Materials:** Created comprehensive teaching materials for Algorithms and C programming courses for second-year undergraduates.

Publications [Google Scholar]

- **LeYOLO: New Scalable and Efficient CNN for Object Detection:** Accepted at CRV 2025 – [arXiv:2406.14239](#)
- **Exploring Superposition in Low-Param Vision Models:** Canadian AI Conference (CAIAC), 2025 – [Publication Link](#)
- **Smart-Viticulture and Deep Learning: Challenges and Recent Developments on Yield Prediction:** Smart Life and Smart Life Engineering, Springer Nature 2025 - [Springer Link](#)
- **Improving Edge-AI Image Classification Through the Use of Better Building Blocks:** CLOSER 2024 - [Conference Paper](#)
- **Machine Learning Forecast of Soybean Yields in South Brazil:** Workshop at IE2022 - [IOS Press](#)
- **Applying Knowledge Distillation on Pre-Trained Model for Early Grapevine Detection:** Workshop at IE2023 - [DOI Link](#)

Technical Expertise

- **Programming Languages:** Python, C, C++ **Technologies & Frameworks:** AWS, Docker, Singularity, PyTorch, Scikit-Learn
- **Core Competencies:** Deep Learning, Computer Vision, Neural Architecture Search, Model Optimization, Real-time Inference, Embedded AI, Agricultural Technology, Research Leadership

References

- **Luiz-Angelo Steffenel:** Professor - ✉ luiz-angelo.steffenel@univ-reims.fr
- **Lucas Mohimont:** Postdoctoral Fellow - ✉ lucas.mohimont@univ-reims.fr
- **Nathalie Gaveau:** Professor - ✉ nathalie.gaveau@univ-reims.fr